Differential Geometry: Assignment 3

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Definition 1. *We call $S \subseteq M$ geodesically convex if for each $p, q \in S$ the following holds:*

- *There is a geodesic $\gamma : [0,1] \to S$ of M from p to q, which is minimizing.*
- *If $\widetilde{\gamma} : [0,1] \to M$ is a minimizing geodesic of M from p to q, then $\widetilde{\gamma} = \gamma$*

Definition 2. *A loop in M is a piecewise smooth curve $\gamma : [a,b] \to M$ such that $\gamma(a) = \gamma(b)$*

Definition 3. *M has trivial holonomy at p if all loops based at p have identity parallel transport*

Definition 4. *M has locally trivial holonomy at p if there is a neighborhood V such that V has trivial holonomy at p*

Claim 1. *If $v, w \in T_pM$ then*

$$lim_{t \to 0} \frac{d_M(\exp_p(tv), \exp_p(tw))}{|t|} = ||v - w||$$

Proof. Consider the following:

- Let $\gamma_t : [0,1] \to M$ be the minimizing geodesic from $q_1(t) = \exp_p(tv)$ to $q_2(t) = \exp_p(tw)$, for $t \in (-\epsilon, \epsilon)$
- We have
$$\gamma(s,t) = \exp(q_1(t), s\widetilde{\gamma}(t))$$
where $\widetilde{\gamma(t)} \in T_{q_1(t)}M$ is the vector of smallest norm such that
$$\exp(q_1(t), \widetilde{\gamma}(t)) = q_2(t)$$
- The map $(s,t) \to \gamma(s,t)$ is smooth by the inverse function theorem
- Let $E(t) = \langle \partial_1\gamma(0,t), \partial_1\gamma(0,t) \rangle$

- Then

$$E''(t) = 2[\langle \partial_1 \partial_2 \gamma(0,t), \partial_1 \partial_2 \gamma(0,t) \rangle + \langle \partial_1 (\partial_2)^2 \gamma(0,t), \partial_1 \gamma(0,t) \rangle]$$

- Now

$$(\partial_1^2 \partial_2 \gamma)(s,0) = (D_s)^2 (\partial_2 \gamma)(s,0) = (D_s D_t \partial_1 \gamma)(s,0)$$

  which equals

$$(R(\partial_1 \gamma, \partial_2 \gamma)\partial_1 \gamma)(s,0) = 0$$

- Hence $\partial_2 \gamma(s,0) = sv + (1-s)w$

□

1 Small geodesic balls are geodesically convex

Consider the following:

- Suppose $\exp_p(B_\epsilon) \subseteq W$, where W is a uniformly δ normal neighborhood of p and $\epsilon < \delta$
- Define $g : M \to \mathbb{R}$ by $g(q) = [d_M(q,p)]^2$
- Define $h : UTB(\frac{\epsilon}{2}, p) \times (-\frac{\epsilon}{2}, \frac{\epsilon}{2}) \to M$ by $(q,v,t) \to \exp_q(tv)$
- Let $K = [\frac{-\epsilon}{4}, \frac{\epsilon}{4}]$
- Define $f = g \circ h$
- Let $U_0 = \{(q,v) \in UTB(\frac{\epsilon}{2}, p) : f''(q,v,t) > 0 \text{ for all } t \in K\}$
- Let $U = \{q \in B(\frac{\epsilon}{2}, p) : (q,v) \in U_0 \text{ for all } v \in T_q M \text{ with } ||v|| = 1\}$

Claim 2. *range $h \subseteq W$*

Proof. Consider the following:

- Let $q_0 = h(q,v,t)$
- $d_M(q_0, q) = |t| < \frac{\epsilon}{2}$
- $d_M(q_0, p) \leq d_M(q_0, q) + d_M(p, q) < \epsilon$
- But we have that $B(\epsilon, p) \subseteq W$

□

Claim 3. *g is smooth on $B(\epsilon, p)$*

Proof. This follows since $g(\exp_p(v)) = ||v||^2$ and $\exp_p$ is a diffeomorphism from B_ϵ to $B(\epsilon, p)$ □

Claim 4. $f(p, v, t) = t^2$

Claim 5. *We have* $f''(a) = Hess_{h(a)}(g)(h'(a), h'(a))$

Claim 6. $Hess_q(g)$ *is positive definite for* $q \in U$

Claim 7. p *is in the interior of* U

Proof. Consider the following:

- U_0 is open by tube lemma
- Choose an orthonormal frame on some neighborhood V_0 of p. We have the diffeomorphism $\Psi : V_0 \times S^{n-1} \to UTV_0$.
- By tube lemma $U = \{q : (q, v) \in \Psi^{-1}(U_0) \text{ for all } v\}$ is open

□

Claim 8. *Suppose* $q_1, q_2 \in U$ *and* $\gamma : [0, 1] \to U$ *is a minimizing geodesic segment from* q_1 *to* q_2*. Then* $g \circ \gamma$ *is convex*

Proof. We have $(g \circ \gamma)''(t) = \text{Hess}_{\gamma(t)}(g)(\gamma'(t), \gamma'(t)) > 0$ □

Claim 9. $V = B(p, \frac{r}{3})$ *is geodesically convex, where* $B(p, r) \subseteq U$ *for some* $r > 0$

Proof. Suppose $q_1, q_2 \in V$ and $\gamma : [0, 1] \to M$ is a minimizing geodesic in M from q_1 to q_2. Then we have:

- $d_M(q_1, \gamma(t)) = t d_M(q_1, q_2) < \frac{2r}{3}$
- $d_M(p, \gamma(t)) \leq d_M(p, q_1) + d_M(q_1, \gamma(t)) < r$
- by a previous claim $g \circ \gamma$ is convex
- so $d_M(p, \gamma(t)) \leq \max\{d_M(p, q_1), d_M(p, q_2)\} < \frac{r}{3}$

□

2 Riemannian isometries

Suppose $f : M \to N$ is a surjective metric isometry with respect to the Riemannian metric. Let $q = f(p)$

Claim 10. *There is an open ball* $B \subseteq \mathcal{E}_p$ *in* T_pM *centered at the origin and a map* $\Psi : B \to \mathcal{E}_q$ *such that*

- $f(\exp_p(v)) = \exp_q(\Psi(v))$ *for all* $v \in B$

- $\Psi(0) = 0$
- Ψ *is a metric isometry*
- *there is a linear isometry* $A : T_pM \to T_qM$ *such that* $Av = \Psi(v)$ *for all* $v \in B$

Proof. We have $d(f(\exp_p(v)), q) = d(exp_q(\Psi(v)), q) = d(\exp_p(v), p)$. Hence $||\Psi(v)|| = ||v||$ □

Claim 11. *There is a neighborhood* V *of* p *such that* $f_{|V}$ *is smooth*

Claim 12. *We have* $(Df)_p = A$

Claim 13. f *is a Riemannian isometry*

3 Flat manifolds

Claim 14. *The following are equivalent*

1. *If* $\gamma_0, \gamma_1 : [0, 1] \to M$ *are smooth curves from* p *to* q *that are path homotopic then they have the same parallel transport*
2. *The restricted holonomy group* $Hol_0(p)$ *is trivial for all* p
3. M *has locally trivial holonomy at every point*
4. $R = 0$

Proof. Consider the following:

- Suppose (1) holds. Since the constant loop has identity parallel transport, (2) also holds
- Suppose (2) holds and let V be a simply connected neighborhood of p. Let γ be a loop based at p in V. Since γ is path homotopic to the constant loop, it has identity parallel transport. Hence (3) holds.
- Suppose (3) holds, so that there is a neighborhood V such that V has trivial holonomy at p. Let $u, v, w \in T_pM$ and define $\Gamma(s, t) = \exp_p(su + tv)$. Applying Theorem 7.11 of Lee, we get $R_p = 0$. Hence (4) holds.

□